

2022-2023 Spring
MAT124 Final
12/06/2023

1. Maximize the function $f(x, y) = xy^2z$ on the sphere $x^2 + y^2 + z^2 = 4$.
2. Sketch the region corresponding to the double integral

$$\int_0^1 \int_{x^{1/4}}^1 e^{y^5} dy dx$$

and evaluate it.

3. Sketch the region R inside the cardioid $r = 1 + \cos \theta$ and outside the limaçon $r = 2 - \cos \theta$, and set up the polar double integral corresponding to the area of the region R .
4. Using a double integral, find the volume of the solid bounded above by the elliptic paraboloid $z = 4 - x^2 - y^2$ and below by the circular region $x^2 + y^2 \leq 2$ in the xy -plane where $x \geq 0$ and $y \geq 0$.
5. Let us consider the frustum of the cone $z = \sqrt{x^2 + y^2}$ between the planes $z = 1$ and $z = 2$.
 - a. Sketch the graph of the frustum.
 - b. Find the surface area of the frustum.
6. Let S be the region in the cylinder $x^2 + y^2 = 1$ bounded above by the plane $z = 4$ and below by the sphere $x^2 + y^2 + z^2 = 1$.
 - a. Using the spherical coordinates, set up (but do not evaluate!) an integral for the volume of the solid S .
 - b. Using the cylindrical coordinates, find the volume of the solid S .

1. Let $g(x, y, z) = x^2 + y^2 + z^2 - 4$ and then solve the system of equations below using the method of Lagrange multipliers.

$$\left. \begin{array}{l} \nabla f = \lambda \nabla g \\ g(x, y, z) = 0 \end{array} \right\} \quad \begin{array}{l} \nabla f = \langle y^2 z, 2xyz, xy^2 \rangle = \lambda \langle 2x, 2y, 2z \rangle = \lambda \nabla g \\ x^2 + y^2 + z^2 - 4 = 0 \end{array}$$

$$y^2 z = \lambda \cdot 2x \quad (1)$$

$$2xyz = \lambda \cdot 2y \quad (2)$$

$$xy^2 = \lambda \cdot 2z \quad (3)$$

$$(1) \& (3) \rightarrow \frac{z}{x} = \frac{x}{z} \rightarrow x^2 = z^2 \rightarrow x = \pm z \quad (4)$$

$$(1) \& (2) \rightarrow \frac{y}{2x} = \frac{x}{y} \rightarrow y^2 = 2x^2 \rightarrow y = \pm\sqrt{2}x \quad (5)$$

$$(4) \& (5) \rightarrow y = \pm\sqrt{2}z$$

Now, use the constraint to find the coordinates. Write y and z in terms of x .

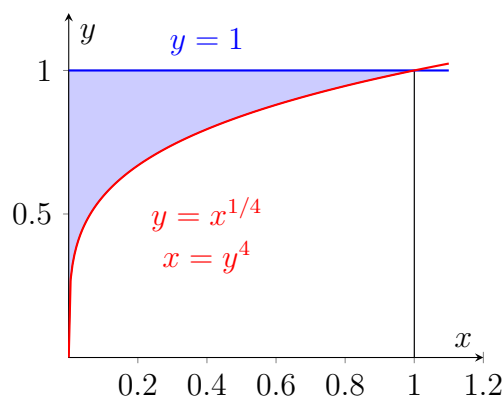
$$x^2 + y^2 + z^2 - 4 = 0 \implies x^2 + (\sqrt{2}x)^2 + x^2 - 4 = 0 \implies 4x^2 = 4 \implies x = \pm 1$$

$$\therefore y = \pm\sqrt{2}, \quad z = \pm 1$$

Evaluate f at either of these points: $(1, \sqrt{2}, 1)$, $(-1, \sqrt{2}, -1)$, $(-1, -\sqrt{2}, -1)$.

$$f_{\max} = f(1, \sqrt{2}, 1) = 1 \cdot (\sqrt{2})^2 \cdot 1 = \boxed{2}$$

2.

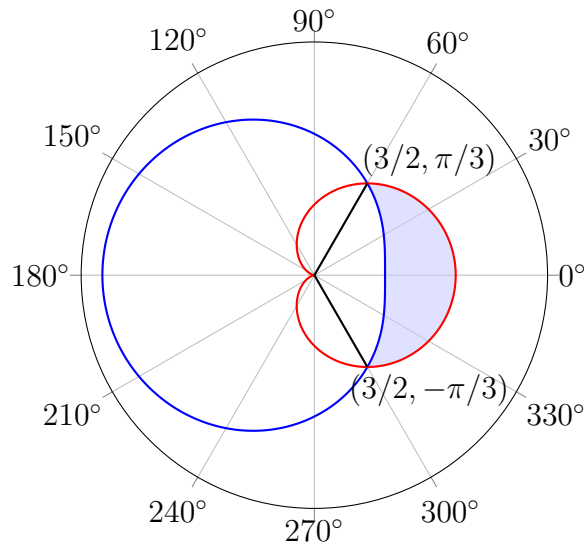


The integral given with this order is difficult to evaluate. Change the order of integration.

$$\int_0^1 \int_{x^{1/4}}^1 e^{y^5} dy dx = \int_0^1 \int_0^{y^4} e^{y^5} dx dy = \int_0^1 y^4 e^{y^5} dy = \left[\frac{1}{5} e^{y^5} \right]_0^1 = \boxed{\frac{e-1}{5}}$$

3. Find where these two curves intersect and then find the area.

$$2 - \cos \theta = 1 + \cos \theta \implies 2 \cos \theta = 1 \implies \cos \theta = \frac{1}{2} \implies \theta = 2k\pi \pm \frac{\pi}{3}, \quad k \in \mathbb{Z}$$



Let A be the area.

$$\begin{aligned} A &= \int_{-\pi/3}^{\pi/3} \int_{2-\cos \theta}^{1+\cos \theta} r \, dr \, d\theta = \frac{1}{2} \int_{-\pi/3}^{\pi/3} [(1 + \cos \theta)^2 - (2 - \cos \theta)^2] \, d\theta \\ &= \frac{1}{2} \int_{-\pi/3}^{\pi/3} [1 + 2 \cos \theta + \cos^2 \theta - (4 - 4 \cos \theta + \cos^2 \theta)] \, d\theta = \frac{1}{2} \int_{-\pi/3}^{\pi/3} (6 \cos \theta - 3) \, d\theta \\ &= \frac{1}{2} \left[6 \sin \theta - 3\theta \right]_{-\pi/3}^{\pi/3} = \frac{1}{2} \left[6 \sin \frac{\pi}{3} - 3 \cdot \frac{\pi}{3} - \left(6 \sin \left(-\frac{\pi}{3} \right) + 3 \cdot \frac{\pi}{3} \right) \right] = \boxed{3\sqrt{3} - \pi} \end{aligned}$$

4. The upper bound is $z = 4 - x^2 - y^2$, while the lower bound is $z = 0$. If we project the domain onto the xy -plane, we see that the upper and lower bounds for y are $\sqrt{2 - x^2}$ and 0, respectively. For x , the integration starts from 0 and ends at $\sqrt{2}$. The volume of the object can be evaluated using the following integral.

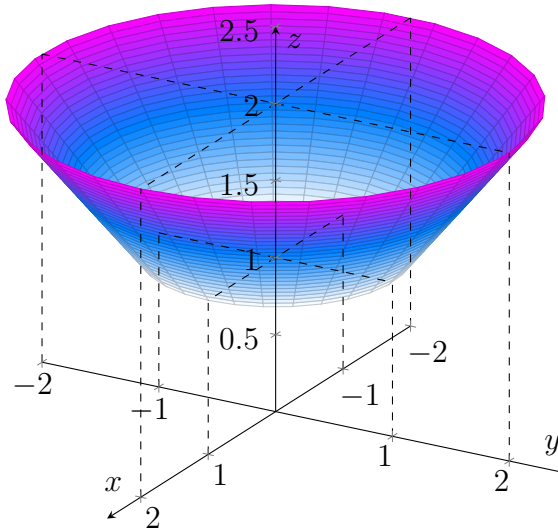
$$I = \int_0^{\sqrt{2}} \int_0^{\sqrt{2-x^2}} [4 - x^2 - y^2 - 0] \, dy \, dx$$

This integral seems a little bit hard. We can switch to polar coordinates for ease.

$$\begin{aligned} x^2 + y^2 = r^2 & \quad \rightarrow \quad \begin{aligned} 0 &\leq z \leq 4 - r^2 \\ 0 &\leq r \leq \sqrt{2} \\ 0 &\leq \theta \leq \pi/2 \end{aligned} \\ dA = dy \, dx = r \, dr \, d\theta & \end{aligned}$$

$$\begin{aligned}
 I &= \int_0^{\pi/2} \int_0^{\sqrt{2}} (4 - r^2) r \, dr \, d\theta = \int_0^{\pi/2} \int_0^{\sqrt{2}} (4r - r^3) \, dr \, d\theta \\
 &= \int_0^{\pi/2} \left[2r^2 - \frac{r^4}{4} \right]_{r=0}^{r=\sqrt{2}} d\theta = \int_0^{\pi/2} 3 \, d\theta = \boxed{\frac{3\pi}{2}}
 \end{aligned}$$

5. a.



b. Using the double integral below, we find the lateral surface area.

$$\begin{aligned}
 A &= \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2} \, dA \\
 &= \iint_D \sqrt{1 + \left(\frac{x}{\sqrt{x^2 + y^2}} \right)^2 + \left(\frac{y}{\sqrt{x^2 + y^2}} \right)^2} \, dA \\
 &= \iint_D \sqrt{1 + \left(\frac{x^2 + y^2}{x^2 + y^2} \right)} \, dA = \iint_D \sqrt{1 + 1} \, dA = \sqrt{2} \iint_D \, dA
 \end{aligned}$$

If we switch to polar coordinates, we can easily evaluate the integral.

$$A = \sqrt{2} \int_0^{2\pi} \int_1^2 r \, dr \, d\theta = \sqrt{2} \int_0^{2\pi} \left[\frac{r^2}{2} \right]_{r=1}^{r=2} d\theta = \sqrt{2} \int_0^{2\pi} \frac{3}{2} \, d\theta = \boxed{3\pi\sqrt{2}}$$

6. a. For spherical coordinates, we have

$$\begin{aligned}
 z &= \rho \cos \phi \\
 r &= \rho \sin \phi \\
 x^2 + y^2 + z^2 &= \rho^2 \\
 dV &= \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta
 \end{aligned}
 \quad \rightarrow \quad
 \begin{aligned}
 x^2 + y^2 &= 1 \rightarrow \rho^2 \sin^2 \phi = 1 \\
 z &= 4 \rightarrow \rho \cos \phi = 4 \\
 z &= \sqrt{1 - x^2 - y^2} \rightarrow \rho \cos \phi = \sqrt{1 - \rho^2 \sin^2 \phi}
 \end{aligned}$$

We have the lower bound for ρ , which is the solution of $\rho \cos \phi = \sqrt{1 - \rho^2 \sin^2 \phi}$.

$$\begin{aligned} \rho \cos \phi = \sqrt{1 - \rho^2 \sin^2 \phi} &\implies \rho^2 \cos^2 \phi = 1 - \rho^2 \sin^2 \phi \\ &\implies \rho^2 (\cos^2 \phi + \sin^2 \phi) = 1\rho^2 = 1 \implies \rho = 1 \end{aligned}$$

However, we have two distinct upper bounds for ρ . We need to find the value of ϕ where the surfaces $\rho \cos \phi = 4$ and $\rho^2 \sin^2 \phi = 1$ intersect.

$$\rho^2 \sin^2 \phi = 1 \rightarrow \rho \sin \phi = 1$$

$$\left. \begin{array}{l} \rho \cos \phi = 4 \\ \rho \sin \phi = 1 \end{array} \right\} \cot \phi = \implies \phi = \cot^{-1}(4)$$

For $\phi < \cot^{-1}(4)$, the upper bound is $\frac{4}{\cos \phi}$. Meanwhile, for $\phi > \cot^{-1}(4)$, it is $\frac{1}{\sin \phi}$. The region in the xy -plane is circular, therefore $0 \leq \theta \leq 2\pi$. As for ϕ , we have $0 \leq \phi \leq \frac{\pi}{2}$. The volume of the object in spherical coordinates can be expressed by

$$V = \int_0^{2\pi} \int_0^{\cot^{-1}(4)} \int_1^{\frac{4}{\cos \phi}} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta + \int_0^{2\pi} \int_{\cot^{-1}(4)}^{\pi/2} \int_1^{\frac{1}{\sin \phi}} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

In fact, we are looking for the minimum value of the upper bound for ρ between $\frac{1}{\sin \phi}$ and $\frac{4}{\cos \phi}$. Hence, we can write the equivalent expression as follows.

$$V = \int_0^{2\pi} \int_0^{\pi/2} \int_1^{\min\left(\frac{4}{\cos \phi}, \frac{1}{\sin \phi}\right)} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

b. For cylindrical coordinates, we have

$$\begin{aligned} z &= z & x^2 + y^2 &= 1 \rightarrow r^2 = 1 \rightarrow r = 1 \\ r^2 &= x^2 + y^2 & & \rightarrow z = 4 \\ dV &= r \, dz \, dr \, d\theta & z &= \sqrt{1 - x^2 - y^2} \rightarrow z = \sqrt{1 - r^2} \end{aligned}$$

The volume can be expressed as follows.

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^1 \int_{\sqrt{1-r^2}}^4 r \, dz \, dr \, d\theta = \int_0^{2\pi} \int_0^1 [z]_{z=\sqrt{1-r^2}}^{z=4} r \, dr \, d\theta \\ &= \int_0^{2\pi} \int_0^1 (4r - r\sqrt{1-r^2}) \, dr \, d\theta = \int_0^{2\pi} \left[2r^2 + \frac{1}{3} (1-r^2)^{3/2} \right]_{r=0}^{r=1} d\theta \\ &= [\theta]_0^{2\pi} \cdot \left[(2 \cdot 1^2 + 0) - \left(\frac{1}{3} + 0 \right) \right] = \boxed{\frac{10\pi}{3}} \end{aligned}$$